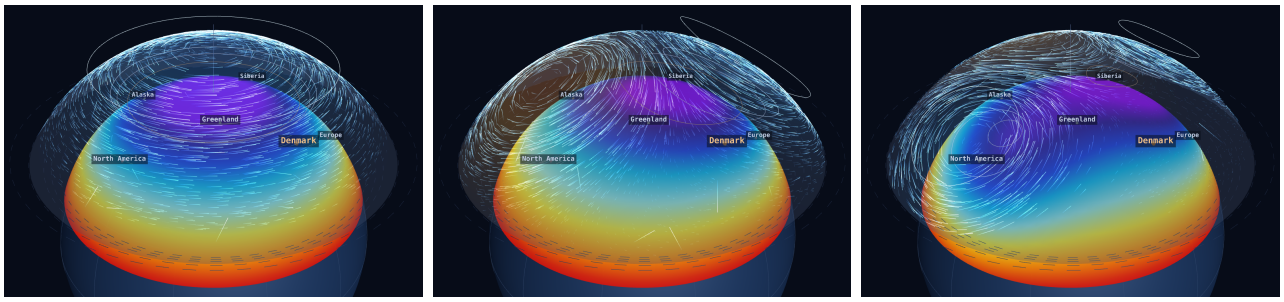
ATMOSPHERIC PHYSICS · METEOROLOGY

The Polar Vortex

Thirty kilometres above the Arctic, a cold whirl of westerly wind circles the pole all winter. In winter 2025/26 it was knocked off course again and again, and in early March the wind reversed completely. Now, in November 2026, the forecasts point to the earliest reversal ever recorded.

Michael Grant · mhgfysik.dk · Updated 5 November 2026



Strong vortex | Displacement | Split

Figure 1. Views from the 3D model at mhgfysik.dk/polar-vortex-en.html. The rings are the vortex edge at five heights (250–2 hPa); the streaks are the wind at 10 hPa and the colour is the temperature. Heights are exaggerated about 60 times. Simplified model, not measured data.

What is the polar vortex?

In winter the polar region gets no sunlight. The Arctic stratosphere cools to about −70 °C, while the air at the same height further south is still warmed by ozone absorbing ultraviolet light. The large temperature difference between pole and mid-latitudes drives a strong westerly wind around the pole. That is the stratospheric polar vortex. The wind is strongest near 60°N, where at 10 hPa it typically blows at 30–50 m/s; it is also called the polar night jet.

The link between temperature contrast and wind is the thermal wind. For a geostrophic wind u in an atmosphere at temperature T,

$$f\frac{\partial u}{\partial z} = -\frac{g}{T}\frac{\partial T}{\partial y}, \qquad f = 2\Omega\sin\varphi.$$

When temperature falls towards the pole ($\partial T/\partial y < 0$), the westerly wind increases with height. The colder the polar night, the stronger the vortex.

We can estimate the height of the 10 hPa level from the barometric formula $z \approx H\ln(p_0/p)$ with scale height $H \approx 7$ km: $z \approx 7\ \text{km} \cdot \ln(1013/10) \approx 32$ km. That is about twice the height of the jet stream that aircraft use, which should not be confused with the polar vortex. The jet stream lives in the troposphere, where the weather happens.

Planetary waves push the vortex

Mountain ranges such as the Rockies and the Himalaya, and the contrast between cold continents and warm oceans, force large stationary waves in the flow: planetary or Rossby waves with only one to three wavelengths around the Earth. They exist because the Coriolis parameter changes with latitude, $\beta = \mathrm{d}f/\mathrm{d}y = 2\Omega\cos\varphi/a$, which at 60°N is about $1.14 \times 10^{-11}\ \text{m}^{-1}\text{s}^{-1}$.

Not every wave can reach the stratosphere. Jule Charney and Philip Drazin showed in 1961 that a stationary wave can only propagate upward if the mean wind $\bar{u}$ is westerly but not too strong:

$$0 < \bar{u} < U_c = \frac{\beta}{k^2 + l^2 + \frac{f^2}{4N^2H^2}}, \qquad k = \frac{s}{a\cos\varphi}.$$

Here s is the wavenumber around the Earth, $N \approx 0.02$ s^{-1} the buoyancy frequency of the stratosphere and l the meridional wavenumber (set to 0 below). With values for 60°N you get $U_c \approx 38$ m/s for $s = 1$, 19 m/s for $s = 2$ and only 10 m/s for $s = 3$. So it is almost only wavenumbers 1 and 2 that get through. Wavenumber 1 displaces the vortex off the pole; wavenumber 2 stretches it and can split it in two. Try it:

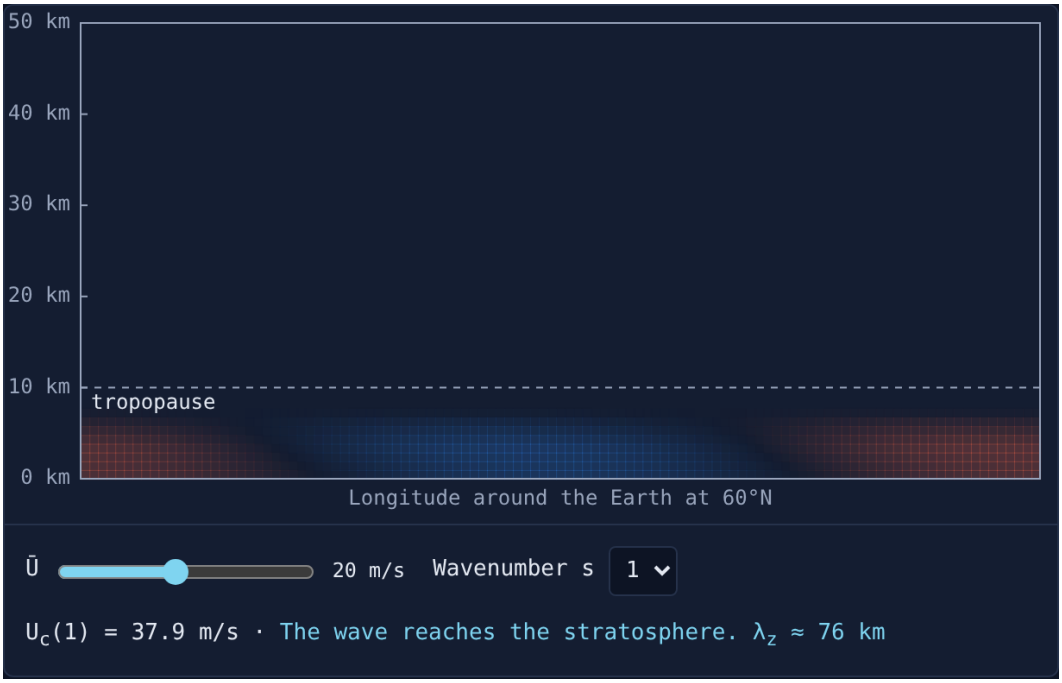


Figure 2. Vertical section along latitude 60°N. Red and blue are high and low pressure anomalies in a stationary planetary wave. When the condition holds, the wave tilts westward with height and its energy (the front) rises into the stratosphere. The amplitude grows with height because the air gets thinner; it is scaled down here. Wavenumber 1, Ū = 20 m/s.

Sudden stratospheric warming

When a strong pulse of planetary waves reaches the stratosphere and breaks, it hands its momentum to the mean flow. That acts as an easterly force that brakes the westerly wind. The upward energy flow of the waves is measured with the Eliassen–Palm flux, and in late February and early March 2026 it showed exactly that: very strong upward wave activity from mid-latitudes into the Arctic stratosphere.

Once the wind slows, the vortex can no longer stay in balance with the cold pole. Air over the pole starts to sink, and sinking air is warmed by adiabatic compression. For dry air potential temperature is conserved, so

$$\frac{T_2}{T_1} = \left(\frac{p_2}{p_1}\right)^{R/c_p}, \qquad \frac{R}{c_p} \approx 0.286.$$

If air sinks from 7 hPa to 10 hPa (about 2.5 km), its temperature rises from 205 K to $205 \cdot (10/7)^{0.286} \approx 227$ K. That matches what was measured in March 2026: the polar cap mean temperature at 10 hPa rose from about 205 K (−68 °C) to about 230 K (−43 °C) within days. Locally the warming can exceed 50 degrees.

A warming is called *major* when the zonal mean wind at 60°N and 10 hPa reverses from westerly to easterly (Charlton & Polvani 2007). On average this has happened about six times per decade since 1958. There are two forms, both shown in the model at the top:

- **Displacement** (wavenumber 1): the whole vortex is pushed towards Eurasia or North America, and a high over the Aleutians takes over the pole.
- **Split** (wavenumber 2): the vortex is stretched and splits into two cores, often one over Canada and one over Siberia or Europe.

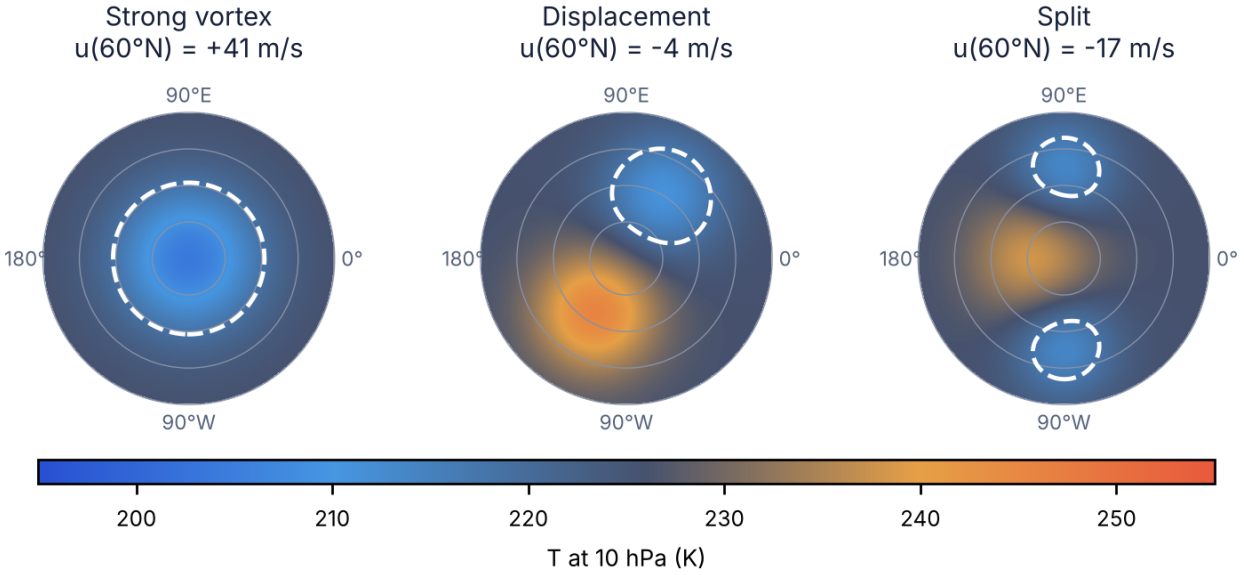


Figure 3. Seen from above the North Pole at 10 hPa in the same model: temperature and the vortex edge (white). The circles are 75°, 60°, 45° and 30°N. In the split the zonal mean wind at 60°N turns easterly.

Winter 2025/26

Winter 2025/26 was unusually restless in the stratosphere. The sequence below is a simplified account of analyses and forecasts in the news coverage; you can play it in the 3D model under *Winter 2025/26*.

Late November – December 2025
A rare, early stratospheric warming weakens the vortex before Christmas. Only a handful of November warmings have been recorded in 70 years.

Mid-January 2026
A major disruption sends Arctic air into North America and Europe, and models show a possible split of the core around 25 January.

Late January – February 2026
Rapid warming over the Arctic. ECMWF ensembles point to a wind reversal and a split around mid-February, and in the Baltic states the February cold was later linked to the split.

Early March 2026
Major SSW: the wind at 60°N and 10 hPa turns easterly, and a 3D analysis from 4 March shows the vortex split into two lobes. The 10 hPa polar cap temperature rises from about 205 K to about 230 K.

Spring 2026
The vortex stays weak and passes into the annual final warming, when it disappears altogether until the autumn.

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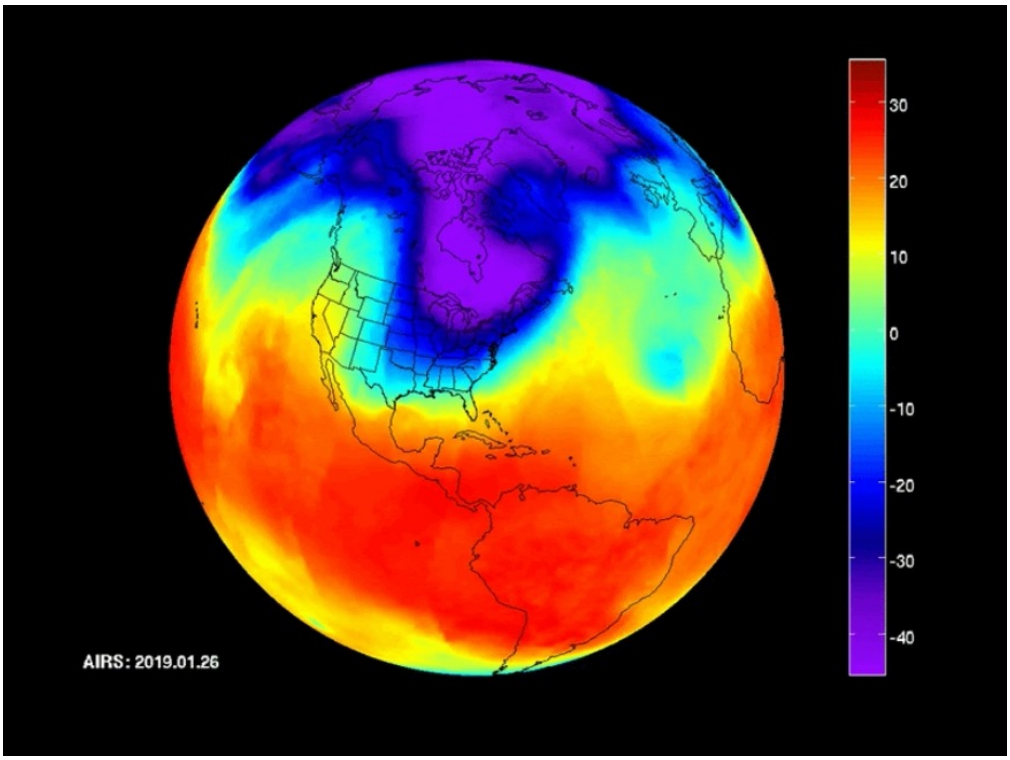
The new vortex has barely formed and the forecasts already show a strong disruption. In late October the ECMWF model showed the 10 hPa wind dropping to record lows for the time of year, with a split into two cores over Siberia and North America and a high over the Aleutians. As of 3 November the reversal had not yet been observed, but about 96% of the European ensemble members showed a reversal by 11 November.

If it happens, it will be the earliest major SSW since records began in 1959. The earliest so far was on 28 November 1968. That does not automatically mean cold weather in November. The Arctic cold reservoir has not built up yet, and any effect at the surface usually comes after a couple of weeks or more, probably in December. At the same time a very strong El Niño pulls the other way.

From 30 km to the weather in Denmark

After a major SSW the disturbance often works its way down through the stratosphere over one to three weeks, as in the model, where the lowest rings react last and weakest. When it reaches the troposphere the jet stream becomes wavier and the North Atlantic Oscillation (NAO) often turns negative. That brings blocking highs over Scandinavia or Greenland and easterly winds over northern Europe. The classic example is the "Beast from the East" in late February 2018, which followed a vortex split on 12 February.

An equally clear example came the next year. On 2 January 2019 the wind at 10 hPa reversed and the vortex split. Three to four weeks later a tongue of Arctic air pushed down over Canada and the northern United States, and Chicago saw about −30 °C on 30–31 January. NASA's AIRS instrument on the Aqua satellite measured the temperature at the ground throughout:



Air temperature at the ground on 26 January 2019 measured by AIRS, in °C. The purple tongue over Canada and the Great Lakes is cold air from the split polar vortex. This is the effect down at the ground; the 3D model at the top shows the cause 30 km up, and the "At the ground" layer shows the same sequence in simplified form. Image: NASA/JPL-Caltech, AIRS Project.

But it does not happen every time. About a third of warmings never reach the weather at all, and the cold can hit North America or Asia just as well as Europe. A weak polar vortex raises the odds of cold weather in our part of the world; it does not guarantee it.

Climate scientists are still debating whether global warming makes the vortex more unstable. Some studies link less sea ice in the Barents and Kara Seas to more disruptions, while others find no clear connection. It is an open research question.

Exercises

1. How high are 300 hPa, 10 hPa and 1 hPa?
Answer: With $z \approx H \ln(1013\,\text{hPa}/p)$ and $H = 7$ km: 300 hPa ≈ 8.5 km, 10 hPa ≈ 32 km, 1 hPa ≈ 48 km. (The scale height varies with temperature, so these are estimates.)

2. Air sinks from 6 hPa to 10 hPa starting at 200 K. What is its new temperature?
Answer: $T_2 = 200\,\text{K} \cdot (10/6)^{0.286} \approx 231$ K, a warming of over 30 degrees with no heat added.

3. Calculate U_c at 60°N for $s = 1$ and $s = 2$, and explain why the summer stratosphere has no planetary waves.
Answer: With $\beta = 1.14 \times 10^{-11}$ m^{-1}s^{-1}, $a\cos\varphi = 3.19 \times 10^{6}$ m, $f = 1.26 \times 10^{-4}$ s^{-1}, $N = 0.02$ s^{-1}, $H = 7$ km and $l = 0$ you get $U_c \approx 38$ m/s and 19 m/s. In summer the stratospheric wind is easterly ($\bar{u} < 0$), so the condition $\bar{u} > 0$ fails and the waves are trapped in the troposphere.

4. Why do major warmings almost never happen in the Southern Hemisphere?
Answer: The Southern Hemisphere has far less land and fewer large mountain ranges, so planetary waves are weaker and the Antarctic vortex is colder and stronger. The only known major SSW in the south was in September 2002; 2019 saw a strong but "minor" warming.

Sources

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